Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
PI Name:	Chui, Talso C. Ph.D.		
Project Title:	Heat Current Effects on the Superfluid Transition		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818-354-3104
Organization Name:	NASA Jet Propulsion Laboratory		
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PI Address 2:			
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City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2000	**End Date:**	10/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Chui, Talso		
COI Name (Institution):	Pearson, David (Jet Propulsion Laboratory) Sukhatme, Kalyani (Jet Propulsion Laboratory)		
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	There is a different way to do work in a Quantum Fluid like superfluid helium. For example, one can move a porous material in a superfluid. The normal component is moved along due to viscous drag, while the superfluid passes through unaffected. Therefore, work is done only to the normal component. These separate behaviors of the superfluid components give rise to a different work term in the thermodynamics, analogous to the work term -PdV of a gaseous system with pressure P and volume V. The research focuses on the rich consequences of this different work term. The objective of this project is to measure thermodynamic quantities such as the superfluid density and the heat capacity when there is a superfluid flow in the system. The superfluid flow can be generated by a heat flux or by pushing the fluid through a constriction that allows only the superfluid component to flow. Another effect due to superfluid flow through a very narrow and short channel is the Jospehson effect, which is an interference effect inside the channel due to the overlaps of the wave functions on each side of the channel. An objective is to understand how the new work term and the fluctuation associated with this new work term affects the physics of the Josephson effect in 4He. The results of the flow measurements will be used to test the predictions of the renormalization group theory, which has recently been extended to cover dynamical situations, such as the superfluid flow being investigated. Depending on the result, the test will either confirm our understanding of dynamical critical phenomena or put into question the adequacy of such theoretical approach. The understanding of the physics of the Josephson effect in 4He can lead to the development of precision gyroscope for space flights and for Earth science studies and for improving the accuracy of real-time GPS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The science being studied lays the foundation for improved GPS with sub-millimeter precision in real-time.
Task Progress:	Last year we made a discovery that the traditional description of how a superfluid gyroscope works is incomplete for superfluid gyroscopes. The description neglected the effects of normal fluid flow in the sensing loop of the gyroscope. To complete the description, we introduced a concept of the frequency-dependent hydrodynamic inductance to take the additional effect into account. With this new concept, we calculated the thermal noise limit and the quantum noise limit of such a gyroscope. We also proposed the idea of using a superfluid gyroscope to improve GPS precision by measuring the Earth's rotational jitter in real time, and correcting its contribution to GPS errors. A preprint describing this has been submitted for publication consideration.
Bibliography Type:	Description: (Last Updated: 01/23/2021)
Articles in Peer-reviewed Journals	Quach HP, Chui TCP. "Low temperature magnetic properties of Metglas 2714A and its potential use as core material for EMI filters." Cryogenics 2004 Jun-Aug;44(6-8):445-9. https://doi.org/10.1016/j.cryogenics.2004.01.006 , Jun-2004